

OpenSmFe: An Open, Processing-Aware and Comparability-Graded Database of Samarium–Iron(–Nitrogen) Permanent-Magnet Properties

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Samarium–iron–nitride ($\text{Sm}_2\text{Fe}_{17}\text{N}_3$) is a leading rare-earth-lean candidate for high-performance permanent magnets, but its published properties are difficult to compare. The nitride decomposes below conventional sintering temperatures, so reported values come from loose powders, aligned powders, bonded magnets and low-temperature consolidated compacts, measured under conditions that papers often leave unstated. Text-mined materials databases now reach tens of thousands of entries, but they record a value without the processing route, sample form and measurement conditions that determine what the value means. We describe OpenSmFe, an open database in which every value is stored as published, converted to SI units, linked to a verbatim evidence sentence, and assigned an explicit A/B/C comparability grade with machine-readable reasons. The schema separates samples (composition, alloy route, milling, melt spinning, annealing, nitriding, consolidation, phase outcome and sample form) from measurements. We apply it to a seed release of 21 author-verified values from 14 samples in 6 open-access papers. No value attains the highest grade. The measurement temperature is unstated for 16 of the 19 values other than Curie temperatures, and the coercivity definition is unstated for 8 of 11 coercivities. The workflow also exposed an internal inconsistency in one source. It caught 1 misattributed and 3 paraphrased evidence statements in the AI-assisted first-draft extraction, all corrected before release. We propose a minimum reporting set for Sm–Fe–N magnet studies and release the data and code under open licenses.

I. INTRODUCTION

Permanent magnets convert between electrical and mechanical energy in electric-vehicle traction motors, wind-turbine generators and a wide range of industrial and defense systems [1, 2]. The highest-performance commercial magnets are based on $\text{Nd}_2\text{Fe}_{14}\text{B}$. Their supply chain is geographically concentrated, and the U.S. Department of Energy lists the rare-earth elements used in them among critical materials [3]. Alternatives that use less of the most supply-constrained rare earths, or that tolerate higher operating temperatures, are therefore of both scientific and strategic interest [2].

$\text{Sm}_2\text{Fe}_{17}\text{N}_3$ is one of the most studied such alternatives. Interstitial nitrogenation of the $\text{Sm}_2\text{Fe}_{17}$ intermetallic converts a compound with planar anisotropy and a low Curie temperature into one with strong uniaxial anisotropy and a substantially higher Curie temperature [4]. The nitride is reported to combine a saturation polarization of about 1.54 T, an anisotropy field of about 14 T and a Curie temperature near 475 °C [5], compared with about 585 K for $\text{Nd}_2\text{Fe}_{14}\text{B}$ [6]. These intrinsic properties make it attractive for magnets that must keep their performance at elevated temperature.

Turning these intrinsic properties into a magnet is the difficulty. $\text{Sm}_2\text{Fe}_{17}\text{N}_3$ decomposes into SmN and α -Fe at elevated temperature [6], so the liquid-phase sintering used for Nd–Fe–B is not available. Practical magnets are instead made by bonding powders in a polymer or metal binder, or by consolidating them at reduced temperature: spark-plasma sintering [6], hot pressing with a low-melting alloy binder [7], or sintering under strictly controlled oxygen [8]. The starting powders themselves are produced by several distinct routes. These include strip casting followed by nitriding and milling [9], calcium reduction–diffusion from oxide precur-

sors [6, 10, 11], and thermal-plasma synthesis of metastable TbCu_7 -type particles [12]. Coercivity in these materials is an extrinsic property governed by microstructure, particle size and surface condition [13]. The processing route is therefore not incidental metadata: it largely determines the measured result.

Comparing published results is correspondingly difficult. Values are reported in CGS and SI units. Magnetization is given per unit mass (emu g^{-1}) or as a polarization (T or kG), and only the latter can be compared directly with a bulk magnet. “Coercivity” may denote the intrinsic coercivity H_{cj} or the normal coercivity H_{cb} , often without saying which. Coercivity depends strongly on temperature [5], so an unstated measurement temperature leaves a value ambiguous. For bulk magnets, standardized hysteresisgraph procedures exist [14, 15]. Most powder studies use open-circuit magnetometry, where sample shape and demagnetization correction matter and are not always reported [16].

Materials-data infrastructure has grown rapidly. Rule-based and machine-learning text mining, exemplified by ChemDataExtractor [17, 18], has produced auto-generated databases such as one of Curie and Néel temperatures [19]. Large language models now extract properties at larger scale and with competitive precision [20]. The Northeast Materials Database (NEMAD) applies them to about 67,000 magnetic materials, including coercivity and remanence [21]. Text mining has also been applied to synthesis procedures [22], and computed-property databases cover intrinsic quantities across hundreds of thousands of compounds [23]. These resources are valuable for discovery and breadth. For an extrinsic quantity such as coercivity, however, a compound–value pair is not sufficient: two values for “ $\text{Sm}_2\text{Fe}_{17}\text{N}_3$ ” can differ by more than an order of magnitude because one was measured on an aligned micrometre powder and the other on a consolidated

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compact. What is missing is a record that keeps the processing context and measurement conditions attached to each value, states openly when they are absent, and lets a reader judge which values may be compared. Keeping that context and provenance attached is also what makes such data findable, accessible, interoperable and reusable [24].

This paper describes OpenSmFe, a database built to supply that record for the Sm–Fe(–N) family. Its contributions are: (i) a schema that separates samples, with their composition and full processing route, from individual measurements; (ii) an explicit, rule-based comparability grade with machine-readable reasons on every value; (iii) an evidence-linked curation workflow in which every value carries a verbatim source sentence and passes automated containment and physical-consistency checks before author verification; (iv) a seed release, analyzed here, that quantifies how often the information needed for comparison is missing; and (v) a proposed minimum reporting set for Sm–Fe–N magnet studies. Sections II and III describe the design and workflow, Sec. IV the seed release, and Sec. V its implications and limitations.

II. DATABASE DESIGN

A. Scope and unit of record

Version 0.1 covers compounds based on the $\text{Sm}_2\text{Fe}_{17}$ structure (the rhombohedral $\text{Th}_2\text{Zn}_{17}$ and hexagonal $\text{Th}_2\text{Ni}_{17}$ types, “2:17”) and the disordered TbCu_7 -type phase (“1:7”), nitrided or not, including substituted variants. The ThMn_{12} -type SmFe_{12} family is deferred to a later version. Soft/hard composites, hybrid Nd–Fe–B/Sm–Fe–N magnets and fluorinated or carbided interstitial compounds are excluded. Only *primary* values are recorded: values a paper reports for its own samples. Values quoted from earlier work, including all values in review articles, are not entered; reviews are used only to locate primary sources.

The unit of record is a *measurement*: one property of one sample at one measurement temperature. Measurements reference a *sample*, which references a *source*. Each table is released as comma-separated values, together with a flat join for spreadsheet users.

B. Sample schema

A sample is one distinct material state described in one paper. Its fields record the nominal composition; the family (2:17 or 1:7); whether it was nitrided; the alloy or powder route (for example arc melting, strip casting, reduction–diffusion, spray pyrolysis, thermal plasma or commercial powder); milling; melt spinning (wheel speed); annealing (temperature, time and atmosphere); nitriding (temperature, time and gas); and consolidation (bonding, sintering, hot pressing or alignment, with conditions). It also records the reported phase outcome, the sample form (powder, aligned powder, powder aligned in resin, sintered or bonded bulk, ribbon or film), whether the sample was magnetically aligned, and the mean particle size. Fields are recorded in the paper’s own terms. Where a paper does not report a field, the record says so explicitly (“not stated in abstract”) rather than leaving a blank that could be read as

zero or as an omission by the curator.

C. Measurement schema and unit harmonization

Each measurement stores the property, the value and unit exactly as printed, a converted SI value, the measurement temperature and its status (stated, implied but unconfirmed, not stated, or not applicable), the instrument when stated, the verbatim evidence sentence and its location in the paper, the comparability grade with reasons, and the verification status.

The properties are the intrinsic coercivity H_{ci} ; the normal coercivity H_{cb} ; a coercivity whose definition the paper does not give, H_c ; the remanence B_r (volume basis) or remanent magnetization σ_r (mass basis); the saturation magnetization; the maximum energy product $(BH)_{\max}$; and the Curie temperature T_C . Coercivities are converted to kA m^{-1} ,

$$H [\text{kA m}^{-1}] = \frac{10^3}{4\pi} H [\text{kOe}] = \frac{\mu_0 H [\text{T}]}{\mu_0} \times 10^{-3}, \quad (1)$$

so that $1 \text{ kOe} = 79.58 \text{ kA m}^{-1}$. Energy products are converted as $(BH)_{\max} [\text{kJ m}^{-3}] = (10^2/4\pi) (BH)_{\max} [\text{MGOe}]$, so that $1 \text{ MGOe} = 7.958 \text{ kJ m}^{-3}$. Polarizations in kG are converted to tesla, and Curie temperatures to kelvin.

Magnetization is kept on its published basis. A specific magnetization σ ($\text{A m}^2 \text{ kg}^{-1}$, numerically equal to emu g^{-1}) corresponds to a polarization

$$J = \mu_0 \rho \sigma, \quad (2)$$

which requires the density ρ of the measured body. Powders and bonded or partially densified magnets are rarely reported with a density, so OpenSmFe does not perform this conversion. Mass-basis values reported for bulk magnets are flagged instead (Sec. IID).

D. Comparability grade

Every measurement receives a grade computed by fixed rules from its own fields (Table I). A *hard* reason makes a value not directly comparable with ambient-temperature data. A *soft* reason leaves it comparable only with care. The grade is

$$g = \begin{cases} C & \text{if } R_{\text{hard}} \neq \emptyset, \\ B & \text{if } R_{\text{hard}} = \emptyset, R_{\text{soft}} \neq \emptyset, \\ A & \text{otherwise,} \end{cases} \quad (3)$$

where R_{hard} and R_{soft} are the sets of applicable reasons, all of which are stored with the row. Room temperature is defined as 290–305 K when stated. An unstated temperature is not assumed to be ambient, because doing so would raise the apparent quality of the record without information to support it. Grades are recomputed on every build. When a verifier records the measurement temperature or coercivity definition from a paper’s methods section, the affected rows re-grade automatically.

E. Consistency checks

Each build runs automated checks and writes a quality report. Converted values must fall within physically plausible

TABLE I. Comparability rules. A value is graded C if any hard reason applies, B if only soft reasons apply, and A otherwise.

Reason	Class
Measured outside 290–305 K (e.g. 10 K)	hard
Measurement temperature not stated	soft
Room temperature implied but not confirmed	soft
Coercivity definition (H_{cj} or H_{cb}) not stated	soft
Mass-basis magnetization reported for a bulk magnet	soft
$(BH)_{\max}$ of a loose powder (depends on assumed packing)	soft
Sample form not confirmed	soft
Value reported inconsistently within the source	soft
T_C approximate, or possibly quoted rather than measured	soft

ranges for the family. Each published number must appear verbatim in its own evidence sentence; this check catches transcription errors and evidence attached to the wrong statement. Within a sample, the energy product must satisfy the ideal square-loop bound

$$(BH)_{\max} \leq \frac{B_r^2}{4\mu_0}, \quad (4)$$

and remanent magnetization may not exceed saturation magnetization. Duplicate identifiers and missing required sample fields are also reported. A release is blocked while any check fails or any field carries an unresolved verification marker.

III. CURATION WORKFLOW

A. Discovery and screening

Candidate papers were identified by a title query of the Directory of Open Access Journals [25] for “Sm₂Fe₁₇N₃”, “Sm-Fe-N” or “Sm₂Fe₁₇”, supplemented by targeted searches for open-access journals. In total 24 records were screened for v0.1:

- 6 were extracted;
- 10 are queued for full-text extraction, because their publishers block automated reading;
- 3 are queued for partial extraction of in-scope reference samples only;
- 2 are reviews, used only to locate primary sources;
- 1 is deferred to the SmFe₁₂ scope;
- 2 were excluded.

Each decision and its reason are released with the data, so the screening can be audited and extended.

B. Extraction

A first draft of each extraction was produced with an AI assistant reading the publisher’s web page. Each value was recorded with its verbatim evidence sentence (at most one sentence) and its location (abstract, section, table or figure caption). For 3 sources the full text was read. For 3 sources only

the abstract could be read, because the publisher’s full-text page was not accessible to automated reading; this is recorded per source. Values that appear only in plots were not digitized in v0.1.

C. Verification and curation corrections

Every first-draft extraction was checked against its source, and all corrections were logged. The curation log records 3 evidence statements that were paraphrases rather than verbatim text and were replaced. In 2 cases the verbatim text showed that a paper reports intrinsic coercivity (“iHc”), and the property was reclassified from H_c to H_{cj} . In 1 case a correct value had been attached to a general statement instead of the sentence reporting the measurement. In 2 cases a measurement temperature had been assumed and was changed to “not stated”. No numerical value was changed during curation. The errors were in evidence, labels and context, which is exactly what a value-only record cannot show. The author then verified all 21 values against their sources before release. Verification was recorded at the release level, as discussed in Sec. V C.

D. Reproducibility

The database is rebuilt from the raw extraction tables by a sequence of scripts that register the inputs with SHA-256 fingerprints, harmonize units, assign grades, apply the verification log, run the checks, and draw the figures. A rebuild from a clean copy reproduces the released tables byte for byte. The tables, figures and every number in this manuscript are generated from the released files [26].

IV. THE SEED RELEASE

A. Composition

Version 0.1 contains 21 measurements of 14 samples from 6 papers published between 2020 and 2025 (Table II). 13 samples belong to the 2:17 family and 1 to the TbCu₇-type family, and 13 of the 14 are nitrided. The measurements comprise 11 coercivities (3 explicitly intrinsic), 5 mass-basis remanent magnetizations, 1 volume-basis remanence, 2 energy products and 2 Curie temperatures. All 21 are listed in Table III. The release is deliberately small: it establishes and tests the schema, grading and workflow before the database is expanded.

B. Coercivity across processing routes

Figure 1 places every coercivity by processing route. At ambient or unstated temperature, values range from 56 kA m⁻¹ (0.056 MA/m) to 1393 kA m⁻¹ (17.5 kOe), a factor of about 25. Because no cross-source value is grade A, this spread should not be read as a ranking of routes. It mixes sample forms, definitions and undocumented conditions.

Contrasts *within* a single source avoid most of these differences, because the instrument, conventions and starting material are shared. Saito [6] reports a coercivity of 12.4 kOe (987 kA m⁻¹) for the reduction–diffusion powder. Spark-plasma consolidation without additives lowers it to 3.05 kOe (243 kA m⁻¹, 25% of the powder value). Adding zinc

TABLE II. Sources in OpenSmFe v0.1. “Text” indicates whether values were extracted from the full text or from the abstract only.

Source	Year	Powder or alloy route	Sample forms	Values	Text
Saito [6]	2024	reduction-diffusion	aligned powder, powder, sintered bulk	11	full
Liang <i>et al.</i> [9]	2023	strip casting	powder	3	abstract
Zheng <i>et al.</i> [10]	2020	ultrasonic spray pyrolysis + H ₂ reduction precursor	powder	1	abstract
Ye <i>et al.</i> [27]	2024	commercial Sm-Fe-N powder	powder	1	abstract
Hirayama <i>et al.</i> [12]	2025	induction thermal plasma	aligned powder in resin	2	full
Zheng <i>et al.</i> [7]	2022	commercial Sm ₂ Fe ₁₇ N ₃ powder	bonded bulk	3	full

TABLE III. All measurements in OpenSmFe v0.1 (each verified against its source). Values as published and converted to SI units; T is the measurement temperature in kelvin (n.s., not stated). Grades follow Table I; reasons for each grade, evidence sentences and full processing fields are in the released files.

Sample	Description	Property	As published	SI	T (K)	Grade
SAITO2024-a	Sm ₂ Fe ₁₇ N ₃ powder (unaligned)	H_c	12.4 kOe	986.8 kA m ⁻¹	n.s.	B
SAITO2024-a	Sm ₂ Fe ₁₇ N ₃ powder (unaligned)	σ_r	75.3 emu g ⁻¹	75.3 A m ² kg ⁻¹	n.s.	B
SAITO2024-b	Sm ₂ Fe ₁₇ N ₃ powder, magnetically aligned (parallel)	σ_r	107 emu g ⁻¹	107 A m ² kg ⁻¹	n.s.	B
SAITO2024-b	Sm ₂ Fe ₁₇ N ₃ powder, magnetically aligned (parallel)	H_c	10.3 kOe	819.6 kA m ⁻¹	n.s.	B
SAITO2024-c	SPS bulk magnet, no additive	σ_r	68.6 emu g ⁻¹	68.6 A m ² kg ⁻¹	n.s.	B
SAITO2024-c	SPS bulk magnet, no additive	H_c	3.05 kOe	242.7 kA m ⁻¹	n.s.	B
SAITO2024-d	SPS bulk magnet, 1 wt% zinc stearate (parallel)	σ_r	90.6 emu g ⁻¹	90.6 A m ² kg ⁻¹	n.s.	B
SAITO2024-e	SPS bulk magnet, 1 wt% zinc stearate + 5 wt% Zn	σ_r	86.8 emu g ⁻¹	86.8 A m ² kg ⁻¹	n.s.	B
SAITO2024-e	SPS bulk magnet, 1 wt% zinc stearate + 5 wt% Zn	H_c	9.82 kOe	781.5 kA m ⁻¹	n.s.	B
SAITO2024-f	Sm ₂ Fe ₁₇ alloy powder	T_C	490 K	490 K	–	B
SAITO2024-g	Sm ₂ Fe ₁₇ N ₃ powder (thermomagnetic, Fig. 3b)	T_C	740 K	740 K	–	B
LIANG2023-a	anisotropic powder, 1.3 μ m	$H_{c,j}$	17.5 kOe	1393 kA m ⁻¹	n.s.	B
LIANG2023-b	anisotropic powder, 3.1 μ m	$(BH)_{\max}$	35 MGOe	278.5 kJ m ⁻³	n.s.	B
LIANG2023-c	anisotropic powder measured at 10 K (particle size not stated in abstract)	$H_{c,j}$	36.1 kOe	2873 kA m ⁻¹	10	C
ZHENG2020-a	USP-HR / R-D powder, 0.616 μ m	H_c	14.7 kOe	1170 kA m ⁻¹	n.s.	B
YE2024-a	anisotropic Sm-Fe-N powder, designed milling	H_c	15.0 kOe	1194 kA m ⁻¹	n.s.	B
HIRAYAMA2025-a	ITP nanopowder x=6.0, nitrided, aligned in resin	H_c	0.056 MA m ⁻¹	56 kA m ⁻¹	300	B
HIRAYAMA2025-a	ITP nanopowder x=6.0, nitrided, aligned in resin	H_c	0.08 MA m ⁻¹	80 kA m ⁻¹	10	C
ZHENG2022-a	hot-pressed magnet, 5 wt% Ce ₇₂ Cu ₂₂ Al ₆ binder	B_r	10.12 kG	1.012 T	n.s.	B
ZHENG2022-a	hot-pressed magnet, 5 wt% Ce ₇₂ Cu ₂₂ Al ₆ binder	$H_{c,j}$	5.63 kOe	448 kA m ⁻¹	n.s.	B
ZHENG2022-a	hot-pressed magnet, 5 wt% Ce ₇₂ Cu ₂₂ Al ₆ binder	$(BH)_{\max}$	21.06 MGOe	167.6 kJ m ⁻³	n.s.	B

stearate and zinc powder recovers it to 9.82 kOe (782 kA m⁻¹, 79%). The same paper reports a remanent magnetization of 107 emu g⁻¹ for magnetically aligned powder measured parallel to the alignment, against 75.3 emu g⁻¹ for unaligned powder. The consolidated magnets range from 68.6 to 90.6 emu g⁻¹. These are mass-basis values for bulk bodies and are graded B, since the corresponding polarization depends on the unreported density [Eq. (2)].

Two sources report coercivity at both ambient and cryogenic temperature. Liang *et al.* [9] give an intrinsic coercivity of 1393 kA m⁻¹ for 1.3 μ m powder and 2873 kA m⁻¹ at 10 K (a factor of 2.1), although the abstract does not say which particle size the 10 K value refers to. Hirayama *et al.* [12] report 56 kA m⁻¹ at 300 K and 80 kA m⁻¹ at 10 K for TbCu₇-type nanoparticles aligned in resin (a factor of 1.4). The 10 K values are graded C. They show how strongly an unstated measurement temperature can affect a coercivity, and why the grade separates them.

C. Energy products and physical bounds

The square-loop bound of Eq. (4) gives a useful sanity check. For the intrinsic polarization of about 1.54 T [5],

the ideal energy product of fully dense, perfectly aligned Sm₂Fe₁₇N₃ is $J_s^2/4\mu_0 \approx 472$ kJ m⁻³. The highest energy product in the release, 35 MGOe (278.5 kJ m⁻³) for 3.1 μ m anisotropic powder [9], is 59% of this ideal. Because it is a powder value, it depends on the packing density assumed in its evaluation and is graded B. For the hot-pressed magnet of Zheng *et al.* [7], the reported remanence of 1.012 T bounds the energy product at 203.7 kJ m⁻³. The reported 167.6 kJ m⁻³ is 82% of that bound, consistent with a well-squared but non-ideal loop. No value in the release violates Eq. (4).

D. Comparability grades and reporting completeness

None of the 21 values is graded A. 19 are graded B and 2 are graded C. The dominant reason is an unstated measurement temperature, which affects 16 of the 19 values other than Curie temperatures (84%). Only 3 of those values carry an explicitly stated temperature. The coercivity definition is not given for 8 of the 11 coercivities (73%). Mass-basis magnetization reported for bulk magnets accounts for 3 further B grades. Removing a single soft reason would raise some rows to A. The 300 K coercivity of Hirayama *et al.*, for example, misses grade A only because the paper does not say whether it is $H_{c,j}$ or H_{cb} .

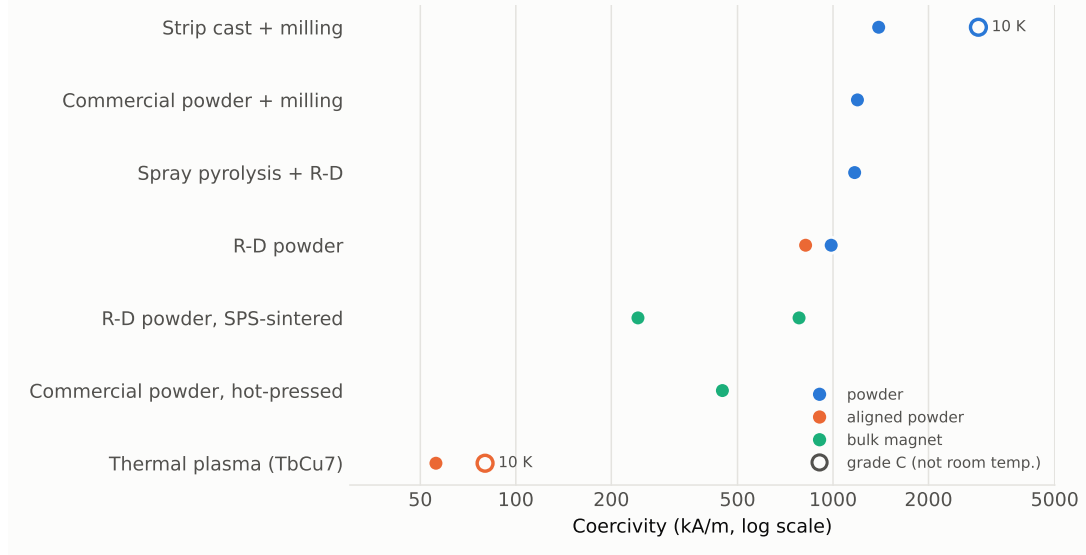


FIG. 1. Published coercivity of Sm-Fe(-N) samples in OpenSmFe v0.1, grouped by powder or alloy route and colored by sample form. Filled markers are values at ambient or unstated temperature (grade B); open markers are values measured at 10 K (grade C), labeled with the temperature. The horizontal axis is logarithmic. Because cross-source values differ in sample form, coercivity definition and documentation, the spread is not a ranking of routes (see text).

Figure 2 summarizes, for each source, the share of its samples for which seven comparison-relevant fields are reported. The overall completeness is 59%. Only 2 of the 6 sources state a measurement temperature for any of their hysteresis values, and 2 state the coercivity definition. Of the 4 sources that nitride their own material (the others start from commercial nitrided powder), the nitriding temperature is stated by 1 in the text available for extraction. The abstract-only sources are marked, since part of this incompleteness may be resolved by reading their methods sections. That is precisely the information the verification step can add, and the grades are designed to rise when it does.

E. Internal inconsistency

One source reports the same quantity with two different values. Zheng *et al.* [7] give a remanence of 10.19 kG in the abstract and 10.12 kG in the results and conclusions, a difference of 0.7%. The database records the value supported by the results section and flags the discrepancy on the row. The difference is small, but a value-only extraction would have recorded whichever figure it encountered first, with no indication that the source disagrees with itself.

V. DISCUSSION

A. Implications for automated extraction

The seed release shows why automated, value-only extraction is not enough for extrinsic magnet properties. The information that decides whether two values can be compared (measurement temperature, coercivity definition, sample form and basis) is missing or implicit for most values, even in recent open-access papers. An automated pipeline that does not record these fields must either discard the value or silently as-

Saito 2024	0%	0%	14%	100%	100%	100%	100%
Liang 2023*	33%	67%	0%	67%	100%	0%	100%
Zheng 2020*	0%	0%	0%	100%	100%	0%	100%
Ye 2024*	0%	0%	100%	0%	0%	0%	100%
Hirayama 2025	100%	0%	100%	100%	100%	100%	100%
Zheng 2022	0%	100%	100%	100%	100%	100%	100%
	Meas. temperature	Hc definition	Nitriding temp.	Particle size	Phase outcome	Alignment	Consolidation

FIG. 2. Reporting completeness by source: the share of each source’s samples (rows) for which a comparison-relevant field (columns) is reported in the text available for extraction. Fields that do not apply (nitriding for samples made from commercial nitrided powder or not nitrided; consolidation for powders) count as reported. Asterisks mark sources read from the abstract only.

sume them. Both choices are invisible to the user. Recording the grade and its reasons makes the uncertainty explicit and auditable. The curation log adds a complementary lesson: the AI-assisted first draft transcribed every number correctly, but 3 of its evidence statements were paraphrases and 1 was attached to the wrong sentence. Verbatim evidence and an automated containment check are inexpensive safeguards against both.

Large text-mined databases and a curated, graded database are complementary. The former provide breadth and discov-

ery [19, 21]; the latter provides a reference against which the accuracy of automated extraction can be measured for a specific materials family. A systematic comparison of OpenSmFe with text-mined entries for the same papers is planned for a later release.

B. Recommended minimum reporting set

The fields that most often prevented comparison suggest a minimum reporting set for experimental Sm–Fe–N magnet studies:

1. the measurement temperature of every hysteresis measurement;
2. the coercivity definition (H_{cj} or H_{cb}) with units;
3. the instrument, maximum applied field, sample geometry and demagnetization correction, following established procedures for bulk magnets where they apply [14, 15];
4. the basis of every magnetization value and, for bulk or bonded magnets, the density;
5. the alloy route and the nitriding temperature, time, gas and pressure;
6. the phase analysis, including any α -Fe fraction;
7. the particle-size distribution and the degree of alignment;
8. key values in a table rather than only in figures.

None of these items is costly to report, and each one, if present, raises the comparability of the corresponding values.

C. Limitations

Version 0.1 is a seed release of 6 papers. It demonstrates the method and does not describe the field; patterns in Fig. 1 reflect which papers were read, not which routes perform best. Only open-access papers are included, so the much larger subscription literature, including foundational work from the 1990s, is not yet represented, and the sample is skewed toward recent years. Of the 6 sources, 3 were read from the abstract only, and values appearing only in figures were not digitized. Values are recorded as published, without correction for demagnetization, density or sample shape. Compositions are nominal. Verification was recorded at the release level: the author verified every value against its source, but the confirmations that the verification log can hold row by row (the measurement temperature and coercivity definition from each paper’s methods section) were not captured for v0.1. The grades are therefore conservative and can only rise as those confirmations are added.

D. Outlook

Version 1.0 will extend the release to the wider open-access literature, including U.S. Department of Energy public-access manuscripts and freely available Japanese journals in which much of the Sm–Fe–N literature appears. Values will be extracted from full-text PDFs and verified in batches, with row-level confirmations of temperature and coercivity definition. The planned composition–process–property maps are coercivity against particle size by route, remanence ratio against

alignment, nitriding conditions against phase outcome, and Curie temperature against substitution. They will be built only from values whose grades allow the comparison.

VI. CONCLUSIONS

OpenSmFe records Sm–Fe(–N) permanent-magnet properties together with the processing route, sample form and measurement conditions that give them meaning. It attaches a verbatim evidence sentence and an explicit comparability grade to every value. Applied to a seed release of 21 verified values, the approach shows that no value can yet be compared across sources without caveats, chiefly because measurement temperatures and coercivity definitions go unreported. It also shows that verbatim evidence and simple automated checks catch errors that value-only extraction would propagate. The schema, grading rules, workflow and data are openly available for reuse and extension, and the proposed reporting set offers a low-cost path to more comparable data in future studies.

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AI assistance—The author used Claude (Anthropic; model Claude Opus 5.5) to search the literature, draft first-pass extractions, write the database code, figures and checks, and draft and revise the text. The author directed each step, verified every database value against its source, approved the scope and grading rules, checked the manuscript, and takes full responsibility for its content.

Data availability—The database, raw extraction tables, curation and verification logs, and the code that rebuilds every table, figure and number in this paper are openly available at <https://github.com/johemeeks/opensmfe> (data CC BY 4.0, code MIT) and archived on Zenodo [26].

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